Ethernet Protocol

What it is

Ethernet is a family of wired LAN networking protocols standardized as IEEE 802.3 that define frame formats, MAC addressing, and the physical/media access rules for transmitting variable-sized packets over shared and switched electrical or optical media. The current standard family spans 10 Mb/s through 400 Gb/s link speeds, with `48-bit` MAC addresses and `1500-byte` default payload.

Why it matters

Every TCP/IP packet, ARP request, and DHCP exchange a developer sees on a local network is encapsulated in an Ethernet frame; understanding the frame layout, MTU, and switch behavior is essential for debugging latency, jumbo frames, VLANs, and NIC offloads.

How it actually works

Frame structure (IEEE 802.3)

The classic Ethernet II / 802.3 frame is a contiguous byte sequence with this exact layout:

Field	Size (bytes)	Purpose
Preamble	`7`	`10101010` × 7 — clock synchronization
SFD (Start Frame Delimiter)	`1`	`10101011` — marks frame start
Destination MAC	`6`	`48-bit` receiver address
Source MAC	`6`	`48-bit` sender address
EtherType / Length	`2`	`≥ 1536 (0x0600)` = EtherType (protocol ID); `≤ 1500` = 802.3 length
Payload	`46`–`1500`	Data (L3 packet); padded to `≥ 46` bytes
FCS (Frame Check Sequence)	`4`	CRC-32 over all fields except preamble/SFD
Interpacket Gap	`12`	`96-bit` idle period between frames

Total on-wire: `64`–`1518` bytes for untagged frames, `68`–`1522` with `802.1Q` VLAN tag (4 extra bytes: `0x8100` TPID + `2-byte` TCI with 12-bit VLAN ID + 3-bit PCP/DEI).

The minimum frame size of `64 bytes` (excluding preamble/SFD) exists specifically so that a collision can be detected by a sender before it finishes transmitting — see CSMA/CD below. The payload minimum of `46 bytes` is derived: `64` − (6+6+2+4) = `46`.

MAC addressing

A MAC address is `48 bits`, written as six hex octets (`AA:BB:CC:DD:EE:FF`). The low-order bit of the first octet is the **I/G (Individual/Group)** flag — `0` = unicast, `1` = multicast. The second-lowest bit is the **U/L (Universal/Local)** flag — `0` = OUI assigned by IEEE, `1` = locally administered. The `FF:FF:FF:FF:FF:FF` broadcast address sets all bits, including I/G. NICs also accept frames addressed to any multicast group they have joined (e.g., IPv4 multicast maps to `01:00:5E:00:00:00` + low 23 bits of the group IP).

CSMA/CD (half-duplex legacy)

Carrier Sense Multiple Access with Collision Detection was the original shared-medium access method on `10BASE5`, `10BASE2`, and early `10BASE-T` hubs:

1. **Carrier sense** — NIC listens; if the medium is idle for `96 bit times` (the interframe gap), it may transmit.
2. **Transmit and listen** — while transmitting, the NIC monitors for voltage/current anomalies indicating another station sent simultaneously.
3. **Collision detection** — on detecting a collision, the NIC transmits a `32-bit` jam signal to ensure all stations see it.
4. **Backoff** — both stations wait a random number of slot times (`512 bit times` at `10 Mb/s` = `51.2 µs`) chosen via truncated binary exponential backoff: after the `i`-th collision, pick `r` uniformly from `[0, 2^k − 1]` where `k = min(i, 10)`; after `16` collisions, the frame is dropped.
5. **Retry** — attempt retransmission up to `16` times.

CSMA/CD is **disabled** on full-duplex links (every modern switch port), because each direction has its own channel and collisions are physically impossible. Modern Ethernet is effectively point-to-point full-duplex.

The 802.3 standard and physical layers

IEEE 802.3 defines a unified MAC and a large family of physical layer (PHY) clauses. Key encodings:

- **10BASE-T** — `10 Mb/s`, Manchester encoding, Cat3/5 UTP, 100m reach.
- **100BASE-TX** (`Fast Ethernet`) — `100 Mb/s`, 4B/5B + MLT-3, Cat5.
- **1000BASE-T** (`Gigabit`) — `1 Gb/s`, PAM-5 over 4 pairs, Cat5e/Cat6.
- **10GBASE-T** — `10 Gb/s`, PAM-16 (DSQ128) over Cat6a/Cat7, 55–100m.
- **100GBASE-** family — `100 Gb/s` using NRZ or PAM-4 on fiber (and later copper), with multiple lane variants like `100GBASE-SR4`, `100GBASE-LR4`, `100GBASE-CR4`.

Each PHY has a defined **bit error rate target of 10^{-12}** for fiber backbones and 10^{-10} for some copper variants, which informs FEC requirements.

Flow control and PAUSE frames

For full-duplex links, 802.3x defines **PAUSE frames** (EtherType `0x8808`, opcode `0x0001`) carrying a `2-byte pause_time` in quanta (one quanta = `512 bit times` at the link speed). The receiving station sends `802.3x PAUSE` to ask the far end to stop for N quanta; symmetric PAUSE pauses both directions. Modern data centers largely replaced this with **Priority Flow Control (802.1Qbb)** which operates per-priority class (8 classes) and is lossless for FCoE/RoCE storage traffic.

Auto-negotiation

Clause 28 / Clause 73 auto-negotiation uses **NLP (Normal Link Pulse)** sequences called **FLPs (Fast Link Pulses)** — 33 pulses in `16` link pulses, encoding a `16-bit` selector + technology ability word. Devices advertise their highest common speed and duplex; both sides pick the highest mutually supported mode. Auto-negotiation is mandatory for `100BASE-TX` and faster twisted-pair PHYs.

Switching and learning

A modern Ethernet switch maintains a **MAC address table (CAM table)** mapping `MAC → port`. On boot this is empty; the switch floods unknown unicast frames out all ports except the ingress. When a frame arrives, the switch (1) records `source MAC → ingress port` with a timestamp, (2) looks up the destination MAC, (3) forwards only to the known port (or floods if unknown/broadcast/multicast). Entries age out after a configurable timeout (typically `300` seconds).

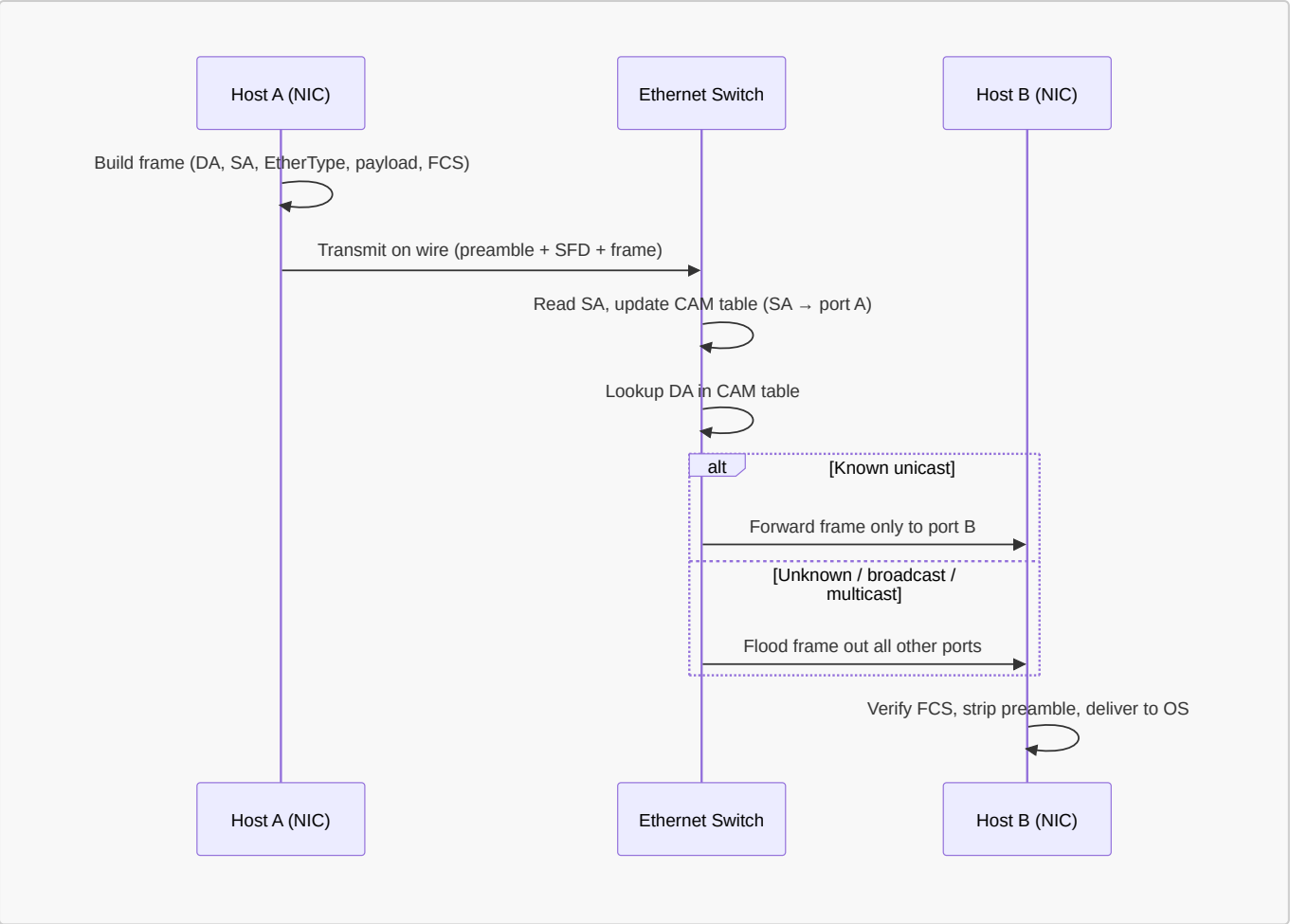
Jumbo frames

Non-standard frames larger than the `1500-byte` standard MTU. Common maximum is **`9000 bytes`** payload (so total frame ≈ `9022` bytes with VLAN tag), used in storage (iSCSI, NFS) and HPC fabrics to reduce per-frame overhead. Both ends and every switch in the path must agree, or frames are silently dropped.

Checksum (FCS)

The trailing `4-byte` FCS is a **CRC-32** with polynomial `0x04C11DB7` (the same one used in ITU-T V.42, MPEG-2, and Gzip's trailing CRC), computed over Destination MAC through Payload. Receivers verify the FCS and discard mismatched frames silently — a corrupted Ethernet frame never reaches the OS; this is why `tcpdump` won't show bad packets unless you capture raw.

Architecture / flow



Key terms

- **MTU** — Maximum Transmission Unit; the largest `payload` (L3 packet) that fits in a single frame. Default `1500` bytes; jumbo frames commonly `9000`.
- **EtherType** — A `2-byte` field identifying the L3 protocol encapsulated in the payload (e.g., `0x0800` IPv4, `0x86DD` IPv6, `0x0806` ARP).
- **PHY** — The physical layer device that handles encoding/decoding between digital MAC and analog electrical/optical signals.
- **FCS** — Frame Check Sequence; a `32-bit` CRC over the frame body used by the NIC to drop corrupted frames.
- **CAM table** — Content-Addressable Memory table inside a switch mapping learned MAC addresses to physical ports.
- **PAUSE frame** — An `802.3x` flow-control frame asking the sender to stop transmitting for N `512-bit` quanta.

Example

A raw 802.3 frame capturing an IPv4 ARP request as seen on the wire (excluding preamble/SFD), showing the exact byte layout:

```
ff ff ff ff ff ff   ← Destination MAC (broadcast)
00 1a 2b 3c 4d 5e   ← Source MAC
08 06               ← EtherType: ARP (0x0806)
00 01               ← Hardware type: Ethernet
```

```
08 00               ← Protocol type: IPv4
06                  ← Hardware size: 6
04                  ← Protocol size: 4
00 01               ← Opcode: request
00 1a 2b 3c 4d 5e   ← Sender MAC
c0 a8 01 0a         ← Sender IP (192.168.1.10)
00 00 00 00 00 00   ← Target MAC (unknown, filled with zeros)
c0 a8 01 14         ← Target IP (192.168.1.20)
[4-byte FCS appended]
```

Demonstrates how an Ethernet header wraps an L3 protocol identified by the `0x0806` EtherType.

Common mistakes

- **Assuming `1500` is the hard MTU limit on modern NICs** — almost all current NICs and switches support `9000`-byte jumbo frames, but they're off by default and per-interface.
- **Forgetting the interframe gap when calculating wire utilization** — back-to-back frames are separated by `12 bytes` of idle, so `10 Gb/s` cannot deliver more than `(line rate) × (1500 / (1500 + 20)) ≈ 98.7%` payload throughput for `1500`-byte frames.
- **Confusing duplex mismatches with cable issues** — a hard-coded `100 Mb/s` full-duplex port connected to an auto-negotiating `1 Gb/s` switch typically ends up at `100 Mb/s` half-duplex with severe late collisions and CRC errors.
- **Treating FCS errors as "physical layer problems" only** — they also occur from NIC offload bugs (especially LRO/TSO) or MTU mismatches where a device sends giant frames that get truncated.

Lesser-known internals

- **The preamble is stripped before the frame is handed to the OS**, so `tcpdump -e` shows `DA, SA, EtherType, payload, FCS` but never the preamble/SFD. The `96-bit` interframe gap is also not visible to software — it is a PHY-level idle pattern.
- **802.1Q VLAN tags change the effective MTU math**: a `1500`-byte IP packet inside an `802.1Q` frame is a `1518`-byte Ethernet frame, and switches enforce the `1522`-byte max tagged MTU. Trunk ports carrying QinQ (`802.1ad`) need `802.1Q` MTU of at least `1530` bytes — `MTU 1500` IP traffic on a QinQ trunk silently drops.
- **NICs offload frame construction**: with **TSO (TCP Segmentation Offload)** the kernel hands the NIC a single `64 KB` super-frame and the NIC chops it into `1500`-byte frames including correct IP/TCP checksums; `ethtool -k <iface>` shows `tcp-segmentation-offload: on`. This is why `tcpdump` running on the *sender* sees the giant pre-segmented frame, while captures downstream see real wire-sized frames.

Topics to explore further

- **ARP (RFC 826)** — how `IPv4 ↔ MAC` resolution actually works and why gratuitous ARP is critical for failover.
- **802.1Q / VLAN tagging** — trunk/access ports, native VLAN PVID, and VLAN hopping attacks.
- **LLDP (802.1AB)** — Layer 2 neighbor discovery used by network management tools.
- **PFC / DCB (802.1Qbb, 802.1Qaz)** — lossless Ethernet priorities used in RoCE and FCoE fabrics.

Learn next

- **ARP** — Direct neighbor protocol; resolving `IP → MAC` is the immediate Layer 2 partner to Ethernet addressing.
- **802.1Q VLANs** — Ethernet's standard tagging scheme, built directly on top of the frame format.
- **Spanning Tree Protocol (802.1D / 802.1w)** — How Layer 2 networks avoid loops while still using Ethernet switches.